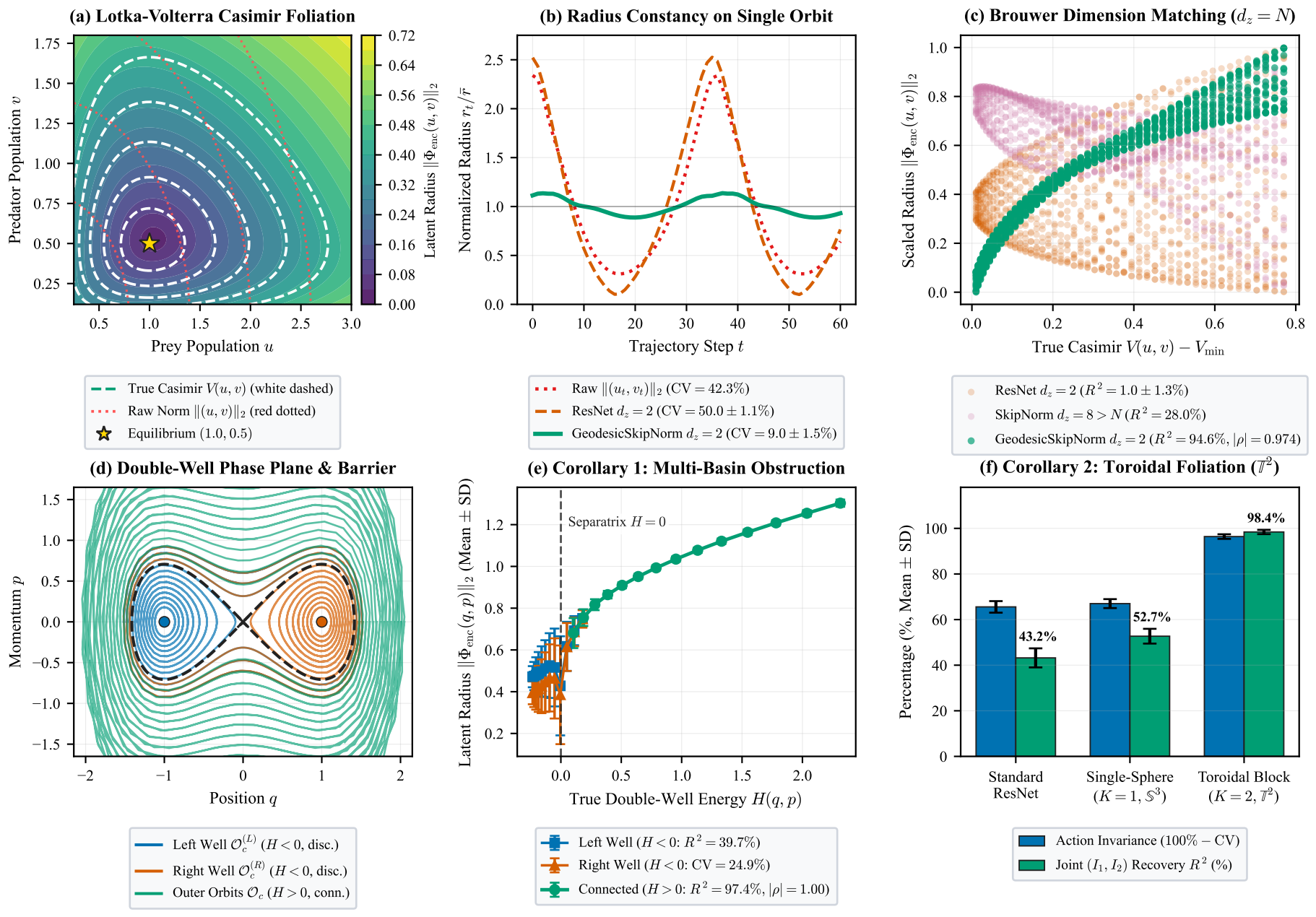

(a) Lotka-Volterra Casimir Foliation
Predator Population v
Prey Population u
Latent Radius $\|\Phi_{\text{enc}}(u,v)\|_2$
True Casimir $V(u,v)$ (white dashed)
Raw Norm $\|(u,v)\|_2$ (red dotted)
Equilibrium (1.0, 0.5)
(b) Radius Constancy on Single Orbit
Normalized Radius $r_t/\bar{r}$
Trajectory Step t
Raw $\|(u_t,v_t)\|_2$ (CV = 42.3%)
ResNet $d_z = 2$ (CV = 50.0 ± 1.1%)
GeodesicSkipNorm $d_z = 2$ (CV = 9.0 ± 1.5%)
(c) Brouwer Dimension Matching ($d_z = N$)
Scaled Radius $\|\Phi_{\text{enc}}(u,v)\|_2$
True Casimir $V(u,v) - V_{\min}$
ResNet $d_z = 2$ ($R^2 = 1.0 \pm 1.3\%$)
SkipNorm $d_z = 8 > N$ ($R^2 = 28.0\%$)
GeodesicSkipNorm $d_z = 2$ ($R^2 = 94.6\%$, $|\rho| = 0.974$)
(d) Double-Well Phase Plane & Barrier
Momentum p
Position q
Left Well $\mathcal{O}_c^{(L)}$ ($H < 0$, disc.)
Right Well $\mathcal{O}_c^{(R)}$ ($H < 0$, disc.)
Outer Orbits $\mathcal{O}_c$ ($H > 0$, conn.)
(e) Corollary 1: Multi-Basin Obstruction
Separatrix $H = 0$
Latent Radius $\|\Phi_{\text{enc}}(q,p)\|_2$ (Mean ± SD)
True Double-Well Energy $H(q,p)$
Left Well ($H < 0$: $R^2 = 39.7\%$)
Right Well ($H < 0$: CV = 24.9%)
Connected ($H > 0$: $R^2 = 97.4\%$, $|\rho| = 1.00$)
(f) Corollary 2: Toroidal Foliation ($\mathbb{T}^2$)
Percentage (%, Mean ± SD)
43.2%
52.7%
98.4%
Standard ResNet
Single-Sphere ($K = 1, \mathbb{S}^3$)
Toroidal Block ($K = 2, \mathbb{T}^2$)
Action Invariance (100% − CV)
Joint (I_1, I_2) Recovery R^2 (%)